Journal of Affective Disorders 241 (2018) 484–491

Contents lists available at ScienceDirect

Journal of Affective Disorders

journal homepage: www.elsevier.com/locate/jad

Review article

The effect of pharmacogenomic testing on response and remission rates in the acute treatment of major depressive disorder: A meta-analysis

Joshua D. Rosenblat[a], Yena Lee[b], Roger S. McIntyre[c,*]

[a] Resident of Psychiatry, Clinician Scientist Stream, University of Toronto, Toronto, Ontario, Canada,
[b] Graduate Student, University of Toronto, Toronto, Ontario, Canada,
[c] Head, Mood Disorder Psychopharmacology Unit, University Health Network, Professor of Psychiatry and Pharmacology, University of Toronto, 399 Bathurst Street, MP 9-325, Toronto, Ontario M5T 2S8, Canada,

ARTICLE INFO

Keywords:
Genetic
Depression
CYP450
Pharmacodynamics
Pharmacokinetics
Antidepressant
Pharmacogenetics
Pharmacogenomics
Cost-effectiveness
Genesight
CNSDose

ABSTRACT

Background: Pharmacogenomic testing has recently become scalable and available to guide the treatment of major depressive disorder (MDD). The objective of the current meta-analysis was to determine if guidance from pharmacogenomic testing results in relatively higher rates of remission and response compared to treatment as usual (i.e., 'unguided' trial-and-error method) in adults with MDD.
Methods: Article databases were systematically searched from inception to December 2, 2017 for human studies assessing the clinical utility of pharmacogenomics in the acute treatment of MDD. Treatment outcomes in MDD may be defined continuously or categorically (i.e., response/remission). Herein, we delimit our focus on categorical outcomes. Using a random-effects model, data was pooled to determine the risk ratio (RR) of response and remission, respectively, in the pharmacogenomic-guided treatment group compared to the unguided group.
Results: Four randomized controlled trials (RCTs) and two open-label, controlled cohort studies were included. The pooled RR for treatment response comparing guided versus unguided treatment was 1.36 (95% confidence interval [CI] = 1.14 to 1.62; $p = 0.0006$; $n = 799$) in favour of guided treatment. The pooled RR for remission was 1.74 (95%CI = 1.09 to 2.77; $p = 0.02$, $n = 735$) also in favour of guided treatment. Heterogeneity in study results suggest that different genetic tests may variably impact response and remission rates.
Limitations: The available evidence is limited, with significant methodological deficiencies.
Conclusion: The current analysis provides preliminary support for improved response and remission rates in MDD when treatment is guided by pharmacogenomics.

1. Introduction

Major depressive disorder (MDD) is a highly prevalent and disabling mental illness affecting more than 350 million people globally as the leading cause of disability worldwide (WHO, 2017). Depression treatment guidelines recommend the use of evidence-based antidepressants in the acute treatment of moderate to severe depression (Kennedy et al., 2016; McIntyre, 2015). Evidence-based treatments are identified through randomized controlled trials (RCTs) assessing the clinical efficacy, safety and tolerability of new treatments. Treatment efficacy can be measured using continuous (i.e., change in depression severity scores, commonly using the Hamilton Rating Scale for Depression [HAMD-17]) or dichotomous variables (i.e., remission and response rates). In clinical studies, response is operationalized as having a reduction of greater than 50% in depression severity scores while remission is operationalized as having a depression severity score within the normal range of non-depressed patients.

Numerous antidepressants are considered 'first-line' with comparable response and remission rates (Cipriani et al., 2018). Unfortunately, however, less than one-third of patients will achieve full remission of depressive symptoms after the first adequate antidepressant trial (Trivedi et al., 2006). After multiple trials of antidepressants from different categories and/or augmenting strategies, approximately one-third of patients still fail to achieve remission of depressive symptoms (Gaynes et al., 2009). The current standard treatment strategy for MDD is sub-optimal and uses a trial-and-error approach until an effective antidepressant or combination treatment providing full remission of depressive symptoms is identified (Cipriani et al., 2018). Towards the aim of personalizing medicine, including patient biotype information may provide a more refined approach to treatment selection, with the expectation of improved safety/tolerability, as well as therapeutic effect.

* Corresponding author.
E-mail addresses: joshua.rosenblat@utoronto.ca (J.D. Rosenblat), yenalee.lee@mail.utoronto.ca (Y. Lee), roger.mcintyre@uhn.ca (R.S. McIntyre).

https://doi.org/10.1016/j.jad.2018.08.056
Received 16 April 2018; Received in revised form 28 July 2018; Accepted 12 August 2018
Available online 14 August 2018
0165-0327/

Pharmacogenomic testing has been proposed and promoted as a point-of-care, scalable tactic to guide treatment selection in MDD (Perlis, 2014). Numerous proprietary pharmacogenomic tests have been developed to guide medication selection by testing for allelic variants of numerous genes that impact the pharmacokinetics and pharmacodynamics of psychotropic medications (as reviewed elsewhere) (Fabbri et al., 2014; Perlis, 2014; Rosenblat et al., 2017). While pharmacogenomics may comport with current theoretical frameworks regarding heterogeneity, tolerability, safety, and treatment outcomes, a data-driven framework would be required before strong assertions can be made, recommending its utility. Conceptually, genetic testing for safety and tolerability needs to be considered separately from genetic testing for efficacy (Goldberg, 2017). Moreover, the health economic implications of pharmacogenomic testing represent another critical endpoint, along with predicting functional and other patient reported outcomes (PROs). Understanding which specific patient sub-population may benefit from testing is also of interest. The current review focuses exclusively on efficacy outcomes, rather than assessing the impact of genetic testing on safety, tolerability, PROs or economic outcomes, which are discussed in other recent articles (Goldberg, 2017; Hornberger et al., 2015; Rosenblat et al., 2017; Zeier et al., 2018).

In a previous systematic review (Rosenblat et al., 2017), extant literature regarding pharmacogenomic testing was found to be of low-quality, with mixed evidence to support augmenting clinical outcomes and/or cost-effectiveness when compared to treatment as usual. Since the publication of the previous review, additional studies evaluating the clinical utility of pharmacogenomic testing have been published, inviting the need for a re-evaluation of the extant literature. Notably, as summarized in the previous review (Rosenblat et al., 2017), the existing studies mostly failed to show a statistically significant difference between guided and unguided treatment, based on the pre-specified primary outcome of change from baseline to endpoint on the HAMD-17. Additionally, most studies did not provide the quantitative data to allow for pooling of effect sizes based on change in HAMD-17 scores. Conversely, most studies consistently report the secondary outcomes of response and remission rates. In the current review, we chose to focus on these important secondary outcomes of relative response (HAMD-17 decreasing by > 50%) and remission (HAMD-17 < 8) rates. The primary objective of this meta-analysis was to determine the effect of pharmacogenomic testing-guided treatment on response and remission rates in the acute treatment of MDD as compared to unguided treatment (i.e., treatment as usual). For a secondary exploratory analysis, we subgrouped pooled samples based on study design, comparing studies using blinded, RCT designs versus un-blinded (i.e., open-label) cohort studies to assess the impact of expectancy bias when participants are made aware that their treatment is being guided by genetic testing.

2. Methods

2.1. Search methods

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Guidelines (Moher et al., 2009) were followed for the current meta-analysis; however, the protocol was not registered prior to conducting the analysis. MEDLINE/PubMed and Google Scholar databases were searched by JDR and YL from inception to December 2, 2017 for published reviews, meta-analyses and primary studies evaluating the impact of pharmacogenomic testing on MDD treatment outcomes. Search terms included various combinations of the following terms: major depressive disorder (MDD), depression, mental illness, mood disorder, antidepressant, response, remission, outcome, pharmacogenetic, pharmacogenomics, pharmacodynamics, pharmacokinetic, genetic testing, genome wide association study (GWAS), CYP450 and personalized medicine. Reference lists from identified articles and ClinicalTrials.gov were also manually searched for additional pertinent references. Google Scholar was used to identify articles that had cited the previously identified pharmacogenomic studies to identify additional potential articles of interest. ClinicalTrials.gov was searched for relevant clinical studies and results were cross-referenced with PubMed and Google Scholar to identify additional pertinent publications.

All identified articles were screened for inclusion in the current systematic review. Two rounds of screening were conducted: Stage 1 - all records from the initial search results were screened based on title and abstract. Pre-clinical articles and/or articles that were clearly outside of the scope of the current review were removed prior to Stage 2 of screening. A low threshold was set to proceed to Stage 2 (i.e., articles proceeded to Stage 2 even if the likelihood for final inclusion was perceived to be relatively low), to maximize sensitivity of the search while disregarding specificity at the current stage. Stage 2 – Full texts of articles identified in Stage 1 were thoroughly reviewed by JDR and YL for inclusion based on the following inclusion criteria. All published prospective human studies, written in English, assessing the effects of utilizing pharmacogenomic testing on improving clinical outcomes of MDD were included (i.e., studies that assessed the efficacy of pharmacogenomic guided treatment). Studies including participants with bipolar depression were excluded. Of note, due to the known limited number of studies, there were no restrictions placed on quality of study, randomization or blinding. As such, studies that were open label, non-randomized or non-blinded were also included. However, we required that all included studies have a control group to allow for the calculation of a risk ratio (RR) for response and remission as this calculation would not be possible in the absence of a comparator arm. Where there was disagreement on inclusion, consensus was reached through discussion between all three authors.

2.2. Data extraction and statistical analysis

Using a standardized data extraction spreadsheet, data was extracted from included studies by two independent reviewers (JDR and YL) to systematically evaluate study characteristics, risks of bias, along with response and remission rates required for calculation of pooled RRs. Where there was disagreement on study quality assessment (i.e., risk of bias), consensus was reached through discussion. Response and remission rates of guided treatment versus unguided treatment at the pre-specified primary endpoint of each study were used for the meta-analysis. Where the response or remission rate was not reported for all or part of the sample, the corresponding author was contacted to obtain these unreported results and given thirty (30) days to respond to the email request.

The pre-specified primary outcome was the pooled RR of response (decrease in HAMD-17 by > 50%) and remission (HAMD-17 reduced to < 8 at primary endpoint) rates comparing guided treatment with unguided treatment. The pooled RRs were also re-analyzed after removal of any outliers (Viechtbauer, 2010). As a secondary outcome, studies were sub-grouped into open-label cohort studies and blinded RCTs and pooled RRs were recalculated for each sub-group.

Determining and pooling RRs and tests of heterogeneity were conducted using Review Manager 5.3 software. We assessed for the presence of influential cases and outliers using R version 3.4.3 and the *metafor* package. Studentized deleted residuals *(t)*, hat *(h)*, and the Mahalanobis distance *(*D) values are reported (Viechtbauer, 2010; Core Team, 2017). A random effects model (using the DerSimonian and Laird method (DerSimonian and Laird, 1986)) was used given expected variability in included study designs as studies included used different study designs and tested different sets of genes along with using different proprietary algorithms to guide treatment. Critical values for pooled RRs were set at 0.05. The overall number needed to treat (NNT) was also calculated using the pooled response and remission rates with the standard equation of 1/(absolute risk reduction). Homogeneity in effect sizes was tested using the Q statistic (Chi^2). Heterogeneity was quantified using the I^2 statistic where 25% = small, 50% = moderate, and 75% = high heterogeneity.

2.3. Assessment of bias

The risk of bias was assessed for all RCTs included in the quantitative synthesis. As per recommendations in the *Cochrane Handbook for Systematic Review of Interventions*, bias was assessed based on the following domains: sequence generation (e.g. based on description of randomization), allocation concealment, blinding/detection bias, attrition bias, reporting bias and other bias. Risk of bias was reported herein if described protocols were concerning for bias in a given domain. Bias and study quality was also systematically assessed in each non-randomized studies using the Newcastle-Ottawa Scale (NOS) for assessing the quality of non-randomized studies (open source NOS tool available at http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp) with a greater number of stars indicative of higher study quality. To assess for publication bias, a funnel plot was created using Review Manager 5.3 software. An Egger's test could not be conducted as a minimum of ten studies is required (Egger et al., 1997) and the current analysis only identified six studies meeting inclusion criteria.

3. Results

3.1. Search results

After removal of duplicates, the initial search yielded 75 records (Fig. 1). After Stage 1 of screening (i.e., reviewing titles and abstracts), 29 full-text articles were evaluated for inclusion. Evaluation of full-text articles yielded four RCTs (Bradley et al., 2018; Perez et al., 2017; Singh, 2015; Winner et al., 2013) as summarized in Table 1. Two open-label cohort studies (Hall-Flavin et al., 2013, 2012) that compared MDD outcomes in guided versus unguided treatment, were also identified, as summarized in Table 2. Identified studies used different proprietary genotyping kits and algorithms to guide MDD treatment, namely, GeneSight (Hall-Flavin et al., 2013, 2012; Winner et al., 2013), CNSDose (Singh, 2015), NeuroIDgenetix (Bradley et al., 2018) and Neuropharmagen (Perez et al., 2017).

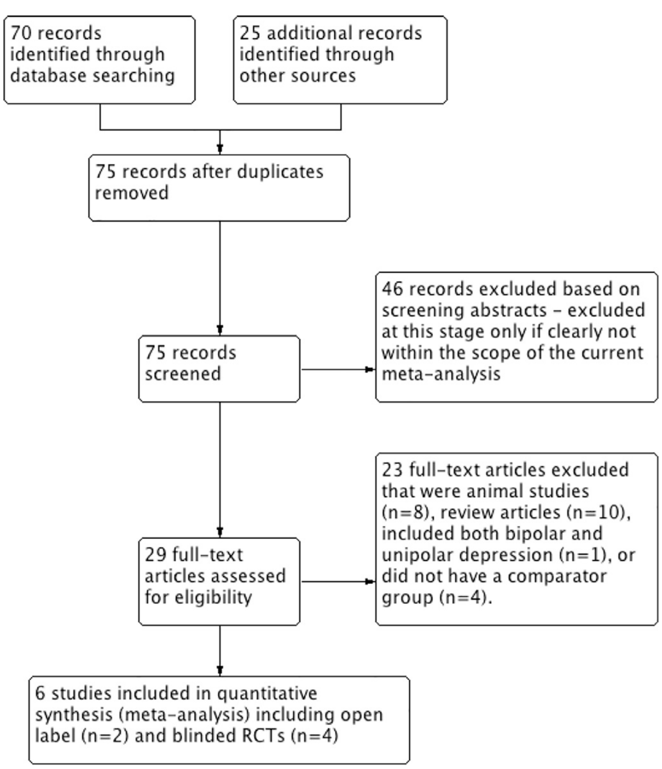

Fig. 1. PRISMA flow diagram for study identification and selection.

3.2. Assessment of study quality and bias

The quality of the included RCTs was assessed systematically in accordance with the *Cochrane Handbook for Systematic Review of Interventions* based on the six aforementioned domains of bias. Overall, the four included RCTs had numerous sources of bias, as summarized in Table 1. All studies were at least partially funded by the companies manufacturing the pharmacogenomic tests. As the prescriber needed to use the pharmacogenomics test report to guide treatment, it was not possible for any of the prescribers to be blinded, representing another possibly significant source of bias. For cohort studies, quality was assessed using the NOS as shown in Table 2. Publication bias was assessed using a funnel plot, as shown in Fig. 2, which was suggestive of a possible publication bias, as studies with larger standard error values were associated with larger effect sizes.

3.3. Pooled effect on response rates comparing guided versus unguided treatment

The RRs of response rates were pooled for all five studies reporting response rates, including cohort studies and RCTs. Notably, the five studies reporting response rates were not identical to the five studies reporting remission rates (of the six total studies included in the meta-analysis, one study only reported remission rates and another only reported response rates). As shown in Fig. 3, based on the results of five studies (n = 799), the overall pooled RR for response comparing guided versus unguided treatment was 1.36 (95% confidence interval [CI] = 1.14 to 1.62; p = 0.0006; n = 799) in favour of guided treatment. Overall, the guided group had a response rate of 50% (201/400) as compared to the unguided group with a pooled response rate of 36% (143/399) with an overall NNT of 7. Heterogeneity was found to be low (Tau^2 = 0.00; Chi^2 = 4.41, df = 4 [P = 0.35]; I^2 = 9%). When analyzing the sub-groups, statistical significance was maintained with blinded RCTs (p = 0.02) and open label cohort studies (p = 0.008); however, unblinded cohort studies showed a greater effect on response rates with an NNT of 6 as compared to an NNT of 8 for blinded RCTs (Table 3).

One study was identified as an influential case, however, did not appear to be an outlier: Perez et al., (2017) ($h = 0.35,\ t = -1.63,\ D = 1.05$). Removing the influential case yielded a less precise estimate of the pooled effect on response rates (COVRATIO = 1.26) and reduced residual heterogeneity (from 9% to 0%). Supplementary Fig. 1 illustrates the hat values plotted against the studentized deleted residuals with point sizes proportionate to the Cook's distances.

3.4. Pooled effect on remission rates comparing guided versus unguided treatment

The RR of remission rates were pooled for all five studies reporting remission rates, including cohort studies and RCTs. As shown in Fig. 4, based on the results of five studies (n = 735), the overall pooled RR for remission rates comparing guided versus unguided treatment was 1.74 (95%CI = 1.09 to 2.77; p = 0.02, n = 735) in favour of guided treatment. Overall, the guided group had a remission rate of 40% (142/353) as compared to the unguided group with a pooled remission rate of 25% (96/383) with an overall NNT of 7. While the overall effect on remission rates was positive, significant heterogeneity was identified (Tau^2 = 0.18; Chi^2 = 14.08, df = 4 [P = 0.007]; I^2 = 72%). As shown in Fig. 4, subgrouping by genetic test showed significant difference with some genetic tests showing a significant positive effect (NeuroIDgenetix, CNSDose) and others showing no significant difference compared to the unguided group (GeneSight, Neuropharmagen).

Two studies were identified as influential cases: Perez et al., (2017) ($h = 0.30,\ t = -2.96,\ D = 1.21$) and Singh et al. (2015) ($h = 0.28,$

Table 1
Double-blinded, randomized controlled trials (RCTs) assessing the clinical utility of pharmacogenomic guided treatment of MDD.

Study	Study design	Study subjects	Results	Outcomes reported	Main study limitations	Bias
Winner et al., 2013	10-week randomized, double-blind (subjects and depression raters), prospective cohort study, assessing clinical utility of GeneSight in an outpatient setting	Adult subjects with DSM-IV diagnosis of MDD with HAMD-17 score > 14 were randomized to a GeneSight guided group ($n = 26$) versus an unguided group ($n = 25$)	At primary endpoint (10 weeks), the guided group had 30.8% improvement in HAMD-17 scores as compared to 20.7% improvement in the unguided group ($p = 0.28$). Response and remission rates trended to favor the guided group, however, also did not reach statistical significance for either response or remission.	-Remission	-Only trend was identified without statistically significant difference -Study may have been underpowered -Treating physician was not blinded	-Fully industry funded
Singh 2015	12-week prospective double-blind (subjects and depression raters), randomized, cohort study assessing the clinical utility of CNSDose®	Caucasian adults with DSM-5 diagnosis of MDD HAMD-17 score > 18 were randomized to a CNSDose® guided group (report guided medication dosing) ($n = 74$) versus unguided group ($n = 74$)	At the primary endpoint (12 weeks), subjects receiving genetically guided prescribing had a 72% remission rate while the unguided group had a remission rate of 28%.	-Remission	-Report guided dosing only, not antidepressant selection -Treating physician was not blinded	-Fully industry funded
Perez et al., 2017	A 12-week, double-blind (subjects and depression raters), parallel, multi-center randomized controlled trial in 316 adult patients with MDD assessing clinical utility of Neuropharmagen guided treatment	Adult subjects with a DSM-IV-TR diagnosis of MDD (both inpatient and outpatient) randomized to unguided ($n = 136$) versus guided treatment ($n = 144$)	No difference in sustained response (primary outcome) was observed (38.5% guided vs 34.4% unguided, $p = 0.5$), but the guided treatment group had a higher responder rate compared to the unguided group at 12 weeks after removing subjects in the guided group when clinicians explicitly reported not to follow the test recommendations (51.3% guided vs 36.1% unguided, $p = 0.0135$). Effects were more consistent in subjects with 1–3 failed drug trials.	-Response -Remission	-Treating physician was not blinded	-Partially industry funded
Bradely et al 2018	A 12-week, prospective, randomized, subject- and rater-blinded approach enrolling 685 patients from clinical providers specializing in Psychiatry, Internal Medicine, Obstetrics & Gynecology, and Family Medicine to assess the clinical utility of NeuroIDgenetix®	Adults age 19 to 87 with a diagnosis of MDD and/or anxiety using the DSM-5 criteria randomized to unguided ($n = 282$) versus guided ($n = 297$) treatment	Study did not report overall changes in depression severity or response/remission rates. Reported no effect for mild depression; however, improved response rates for moderate to severe depression ($p = 0.001$) and improved remission rates in severe depression ($p = 0.02$).	For sub-population: -Response -Remission	-Treating physician was not blinded	-Fully industry funded -Very significant reporting bias: selectively reporting results for subgroups rather than reporting outcomes for the entire sample (Note: authors were contacted and were unable to provide results).

Table 2
Unblinded, open-label cohort studies assessing the clinical utility of pharmacogenomic guided treatment of MDD.

Study	Study design	Study subjects	Results of primary outcome	Results reported	Newcastle-Ottawa Scale (NOS)	Main study Limitations/Bias
Hall-Flavin et al., 2012	8-week non-randomized, open label, prospective cohort study assessing clinical utility of GeneSight in an outpatient setting	Adult (18–75) subjects with DSM-IV diagnosis of MDD with HAMD-17 score > 14 were non-randomly allocated into GeneSight guided group (n = 22) versus an unguided group (n = 22)	At primary endpoint (8 weeks) 18.2% reduction in HAMD-17 ratings for subjects in the unguided group compared with a 30.8% reduction for subjects in the guided group (P = 0.04).	-Response	Selection- ★★★ Comparability (Comp.)- 0 Outcome-★	-Open label, non-randomized study design -Partially industry funded -Reporting bias (did not report remission rates)
Hall-Flavin et al., 2013	8-week non-randomized, open label, prospective cohort study assessing clinical utility of GeneSight in an outpatient setting	Adult (18–72) subjects with DSM-IV diagnosis of MDD with HAMD-17 score > 14 were non-randomly allocated into GeneSight guided group (n = 114) versus an unguided group (n = 113)	At primary endpoint (8 weeks) 46.9% reduction in HAMD-17 scores in the guided group as compared to 29.9% reduction among the unguided group (P < 0.0001).	-Response -Remission	Selection- ★★★ Comparability (Comp.)- 0 Outcome-★	-Open label, non-randomized study design -Partially industry funded

Abbreviations: Confidence Interval (CI), Diagnostic and Statistical Manual of Mental Disorders (DSM), Hamilton Rating Scale for Depression (HAMD), Major Depressive Disorder (MDD)

$t = 1.42$, $D = 0.76$). One study had a studentized deleted residual larger than + /- 1.96 (Perez et al., 2017) (Supplementary Fig. 2). Removal of this outlier leads to a pooled estimate RR of 2.12 (95%CI = 1.54 to 2.91; $p < 0.00001$, $n = 211$; COVRATIO = 0.6025) for remission rates and a reduction in the estimated I^2 value (from 72% to 10%).

4. Discussion

4.1. Brief summary

The current meta-analysis identified six studies that reported response and/or remission rates in the acute treatment of MDD, comparing treatment guided by pharmacogenomic testing with unguided treatment as usual. Quantitatively pooling the results of all identified studies, guided treatment was found to improve response and remission rates in the acute treatment of MDD with an NNT of 7 for both response and remission. For response rates, the observed effects were greater in open-label cohort studies when compared to blinded RCTs; however, both subgroups independently had statistically significant results in favour of guided treatment.

4.2. Explanation of findings

These results are in accordance with our initial hypothesis that an overall effect in favour of guided treatment would be observed given the theoretical benefits of knowing patient-specific pharmacodynamics and pharmacokinetic profiles when prescribing psychotropic medications (Perlis, 2014). We also hypothesized that the effect would be less robust when participants were adequately blinded and randomized. We hypothesized that the improved outcomes would be largely driven by open-label studies as the improved outcomes observed in these unblinded studies may largely be mediated by an enhanced placebo effect, insofar as participants are aware that their treatment is being guided by genetic testing, with an expectancy bias that the 'perfect' personalized treatment was being selected (Rutherford et al., 2010).

Notably, the largest blinded RCT testing the clinical utility of pharmacogenomic testing in MDD was recently completed and presented at the *American Psychiatric Association (APA) General Annual Meeting* (ClinicalTrials.gov: NCT02109939; APA 2018 poster P5-110) and had a negative primary outcome, in spite of being adequately powered with 1200 participants with MDD enrolled (*unpublished*). In this study, the primary outcome of change in HAMD-17 scores from baseline to week-8 of treatment in participants receiving GeneSight-guided therapy compared to those receiving treatment as usual failed to reach statistical significance ($p = 0.1$); however, improved remission rates were observed in the GeneSight-guided group compared to treatment as usual. Of note, these results have not yet been published in a peer-reviewed journal and as such could not be included in the current meta-analysis.

In addition to the overall findings described, significant differences were observed in efficacy when comparing different genetic tests. Of note, however, the current meta-analysis was not designed to compare efficacy between tests, but rather to assess the overall effect of all pharmacogenomic guided treatments. Given the proprietary nature of most of the algorithms used by these genetic tests to guide treatment, it is difficult to discern which specific genes or pharmacogenomic approaches are mediating the improved effects with specific genetic tests as compared to others. For example, CNSDose was shown to have the greatest effect on improving remission rates (guided = 72% versus unguided = 28%); however, this study reported only that a "proprietary pharmacokinetic polygene pathway pharmacogenetic interpretive formula" was used to guide treatment (Singh, 2015). Nevertheless, integrating genetic data from a combination of sources, such as CYP450 genes, serotonin transporter and receptor genes and p-glycoprotein genes likely increases the predictive power of testing, possibly

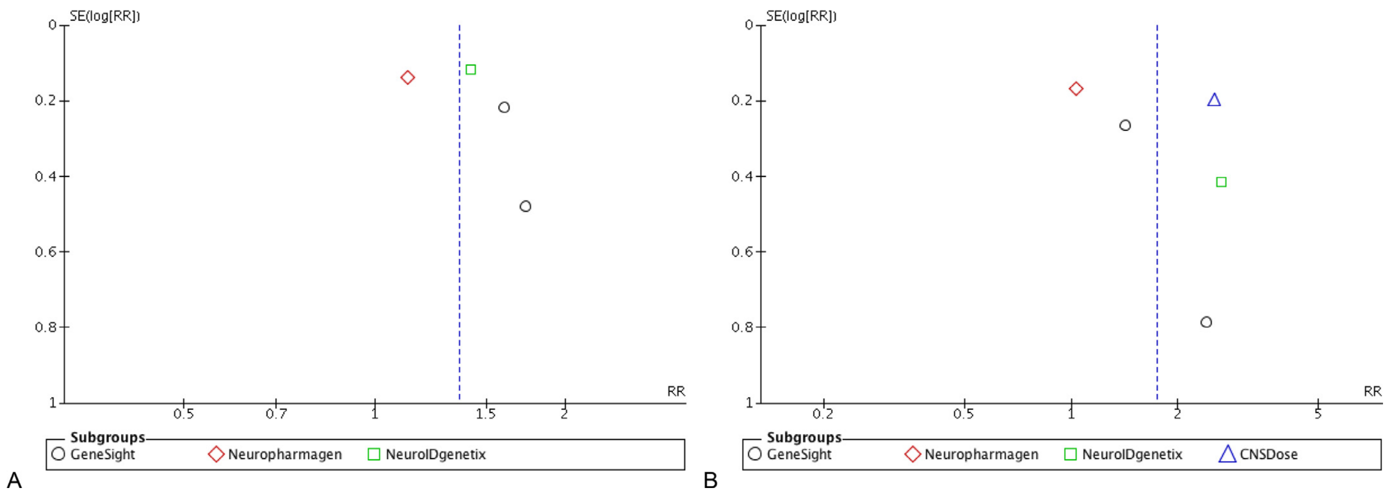

Fig. 2. Funnel plots to assess for publication bias for (A) response rates and (B) remission rates. Standard error (SE), risk ratio (RR).

translating to further improve clinical outcomes in these studies (Altar et al., 2015).

4.3. Limitations

There are several methodological limitations that affect inferences and interpretations that may be drawn from the current study. The small number of studies and inclusion of open-label cohort studies decrease the reliability of the pooled effects reported. We attempted to mitigate this limitation by also reporting subgroup results of RCTs and cohort studies as shown in Table 3. However, the included blinded RCTs had several potential sources of bias as well. Another significant limitation to the reliability of our results was that all studies were at least partially funded by the companies manufacturing the pharmacogenomic tests. Another limitation was the significant heterogeneity of results of included studies when pooling remission rates, suggesting that it may be inappropriate to combine the results of studies using different genetic testing protocols. As such, the generalizability of the current results may be limited to specific genetic tests rather than being generalizable to all pharmacogenomic testing guided treatment algorithms.

5. Conclusion, clinical implications and future direction

Taken together, the current meta-analysis suggests that treatment outcomes in MDD, as proxied by response and remission rates, may be improved by pharmacogenomic testing guidance as compared to treatment as usual. Given the differences in results of blinded versus unblinded studies (i.e., cohort studies), however, it remains unclear if these improved clinical outcomes are mediated biologically via optimization and personalization of prescribing patterns or if the improvement is primarily an epiphenomenon of an enhanced placebo effect when patients know they are receiving 'personalized' treatment (Rutherford et al., 2010). However, the difference in results of blinded

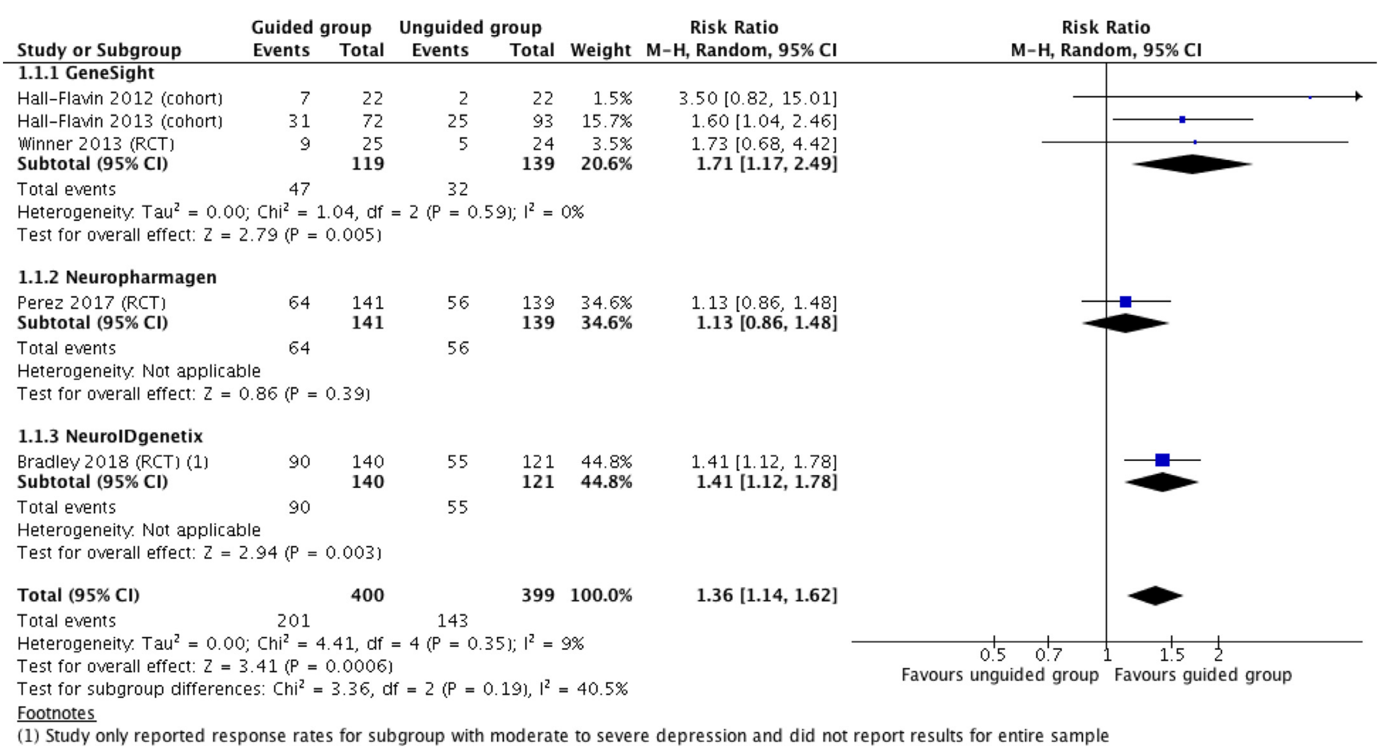

Fig. 3. Pooled risk ratio (RR) of response rates comparing pharmacogenomic guided treatment versus unguided treatment (i.e., treatment as usual).

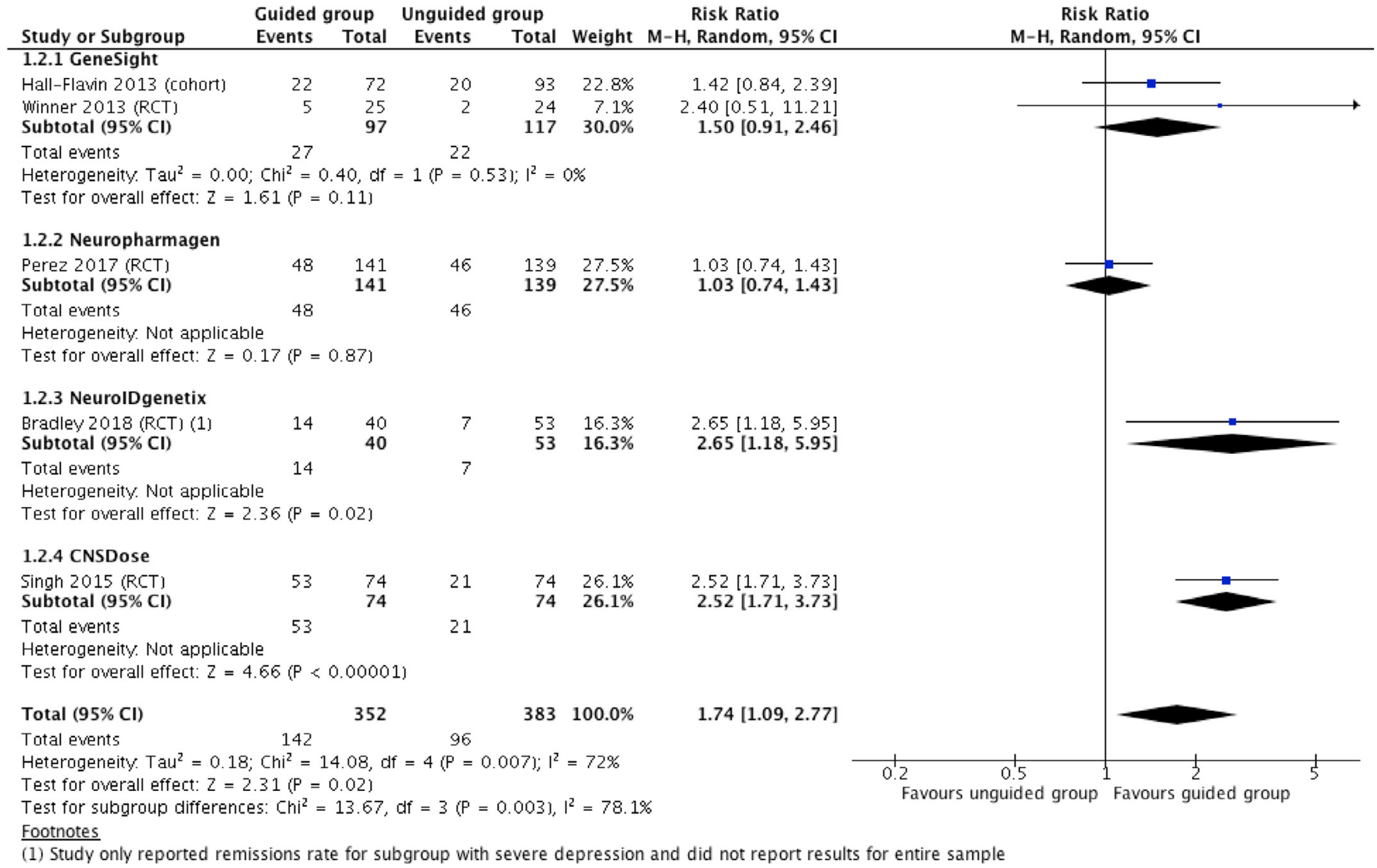

Fig. 4. Pooled risk ratio (RR) of remission rates comparing pharmacogenomic guided treatment versus unguided treatment (i.e., treatment as usual).

versus un-blinded studies was found to be minimal (Table 3), suggesting that the enhanced placebo effect in unblinded studies is likely minor. Nevertheless, to obtain more accurate and meaningful results, expectancy bias must be controlled for, further emphasizing the importance of future well designed, adequately blinded RCTs, to replicate and confirm the observed effects. Future studies should also control for proportions of specific drugs being prescribed as algorithms guiding treatment might preferentially select newer agents with less pharmacogenomic interactions, potentially leading to the guided group having a greater proportion of patients receiving 'better' drugs (i.e., newer, potentially more efficacious and tolerable agents).

The current meta-analysis also found significant heterogeneity in study results, suggesting potential differences in clinical utility of various pharmacogenomic tests so a 'class effect' of pharmacogenomic tests in general should not be assumed. Indeed, as numerous new 'me too' pharmacogenomic tests come to market, improved clinical outcomes with guided treatment should not be assumed, but rather must be validated through well designed, blinded RCTs. The significant differences in pharmacogenomic tests (i.e., differences in genes tested, algorithms used to guide treatment and effects on response/remission rates) highlight the importance of evaluating each test independently and not assuming a 'class effect' for pharmacogenomic guided treatment; evaluation and clinical recommendations must be test specific. For a *specific* pharmacogenomic test to be clinically recommended, the same standard criteria used for drug approval should be met; namely, completion of two positive, well designed, phase III RCTs. The current meta-analysis only identified one RCT per pharmacogenomic test. As such, no pharmacogenomic test had replicated findings to support improved therapeutic efficacy. Therefore, currently available evidence is insufficient to support routine pharmacogenomic testing to guide the treatment of depression.

As the accurate clinical utility of specific pharmacogenomic tests is further determined through independently replicated studies, an evaluation of cost-effectiveness will also be needed to justify public payer reimbursement. Furthermore, the relevant patient subpopulations that may benefit from genetic testing remains unknown and there continues

Table 3
Pooled response and remission rates with subgroup analysis. *Study only published small subset of results, showing remission rates for subgroup with severe depression and response rates for subgroup with moderate to severe depression. **Note: overlap is likely present between remission and response (i.e., participants meeting criteria for remission likely also met criteria for response), however, degree of overlap could not be determined.

Genetic test	Response rates**				Remission rates**			
	Guided	Unguided	*p*-value	NNT	Guided	Unguided	p-value	NNT
Genesight	39% (47/119)	23% (32/139)	**0.005**	6	27% (27/97)	19% (22/117)	0.11	11
Neuropharmagen	45% (64/141)	40% (56/139)	0.39	20	34% (48/141)	33% (46/139)	0.87	105
NeuroIDgenetix*	64% (90/140)	45% (55/121)	**0.003**	5	35% (14/40)	13% (7/53)	**0.02**	5
CNSDose	Not reported				72% (53/74)	28% (21/74)	**< 0.00001**	3
Combined results (full sample)	50% (201/400)	36% (143/399)	**0.0006**	7	40% (142/353)	25% (96/383)	**0.02**	7
Combined results (cohort studies only)	40% (38/94)	23% (27/115)	**0.008**	6	30% (22/72)	22% (20/93)	0.19	11
Combined results (blinded RCTs only)	53% (163/306)	41% (116/284)	**0.02**	8	33% (120/280)	25% (76/290)	0.05	6

to be controversy whether genetic testing should be reserved for patients with treatment resistance/sensitivity or if the benefit would be greatest if testing is done prior to initiation of the first antidepressant (Zeier et al., 2018). Additionally, future studies may include more ethnically diverse samples (rather than primarily participants of European ancestry) as specific ethnicities may benefit less or more from testing. Lastly, future studies should also include PROs rather than delimit to conventional symptomatic efficacy outcomes (e.g. HAMD scores) as PROs are being increasingly recognized as important outcomes to consider with any new treatment or test in psychiatry. For example, PROs might show improved patient satisfaction with genetic testing through improved safety and tolerability, even in the absence of improved efficacy.

6. Declaration of interests

JDR and YL have no competing interests to declare. RSM has received research grant support from Lundbeck, JanssenOrtho, Shire, Purdue, AstraZeneca, Pfizer, Otsuka, Allergan, Stanley Medical Research Institute (SMRI); speaker/consultation fees from Lundbeck, Pfizer, AstraZeneca, Elli-Lilly, JanssenOrtho, Purdue, Johnson & Johnson, Moksha8, Sunovion, Mitsubishi, Takeda, Forest, Otsuka, Bristol-Myers Squibb, and Shire.

7. Contributors

JDR and RSM conceived the project and protocol. JDR and YL performed the literature search and data analysis. JDR wrote the initial draft of the manuscript. YL and RSM provided edits and revisions. All authors read and approved the final submission.

8. Funding source for current study

None

Acknowledgements

None

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jad.2018.08.056.

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